10	A
11	B
12	C
13	D
14	E
15	F

8421

(1) DES =

(a) Permutation :-

Q => 0F0A 0000 0B00 00C1

1100 C	2 0011 00	1101 D	1111	0100 4	1111
1-4	5-8	9-12	13-16	17-20	21-24
0000	1111	0000	1010	0000	0000
1101 D	1111	1111	1111	1111	1111
25-28	29-32	33-36	37-40	41-44	
0000	0000	0000	1011	0000	
1111	1111	0011 02	1111	0011 02	
45-48	49-52	53-56	57-60	61-64	
0000	0000	0000	1100	0001	

C2D040D00000020 2

(b) S-Box =

Q => 1 0 1 1 0 1

421 → 011 → Row → 3

8421 → 0110 → Column → 6

→ Search Row 3, Column 6 in S-box table

=> $\boxed{0001}$

- Expansion Permutation =

EP-box table:-

1	32	01	02	03	04	05
2	04	05	06	07	08	09
3	08	09	10	11	12	13
4	12	13	14	15	16	17
5	16	17	18	19	20	21
6	20	21	22	23	24	25
7	24	25	26	27	28	29
8	28	29	30	31	32	01

Q.) 1 4 5 8 9 12 13 16 17 20 21 24 25 28 29 32

1111 0000 1010 1010 1111 0000 1010 1010

1 2 3 4 5 6 7 8

1. 011110
2. 100001
3. 010101
4. 010101
5. 011110
6. 100001
7. 010101
8. 010101